

Seismic Depth Imaging for O&G Exploration Geoscientists.

Course Provided by Seistech Imaging Inc.

Instructors.

- Dr. Julian Cabrera (Geophysicist, Canada).
- Dr. Ian Jones (Geophysicist, UK).
- Prof. Juan Soto (Structural Geologist, Spain).

Objectives.

- Provide a solid understanding and practical knowledge of seismic depth imaging concepts, methods, and workflows.
- Apply structural geology concepts and methodology to depth velocity model building and seismic imaging.
- Empower personnel to describe the importance of the synergy between geology and geophysics for maximizing the veracity of depth velocity models and seismic images.
- Increase personnel effectiveness in supervising seismic depth imaging projects within the context of exploration and interpretation work.
- Build personnel confidence in evaluating the quality of seismic depth images and velocity models, identifying and discussing potential issues, tests and remedial approaches.

Target Audience.

- Exploration geologists and geophysicists working with seismic imaging products.
- O&G professionals who may lead Exploration teams and become responsible for seismic imaging projects.
- Technical personnel from seismic contractors.
- University students and recent graduates pursuing careers in O&G exploration.

Given that the course aims to provide a solid understanding of seismic depth imaging and its structural geology applications for practising geoscientists, it will not cover advanced theory nor mathematical details, neither will it deal with converted mode or VSP data.

Prerequisites.

- Experience in seismic interpretation for oil and gas exploration.
- Basic understanding of seismic imaging, data processing and structural interpretation.
- Basic knowledge of linear algebra and calculus (for Tomography & FWI sections).

Duration.

- 40 hours.

Course Content

1. Introduction - *Dr. Cabrera, Dr. Jones, Prof. Soto*

- Objectives of seismic depth imaging in Exploration.
- Geology and Geophysics in seismic imaging.
- Course sections overview.
- What the attendees should attain by the end of this course.

2. Acoustic and elastic wave propagation – *Dr. Cabrera*

3. Anisotropy – *Dr. Jones*

4. Seismic illumination – *Dr. Cabrera*

5. Migration algorithms – *Dr. Cabrera*

- Kirchhoff migration.
- Reverse time migration (RTM).

6. Velocity inversion methods, practical aspects and data preparation.

- Ray-based tomography. – *Dr. Cabrera*
- Full waveform inversion (FWI). – *Dr. Jones*

7. Geological imaging models – *Prof. Soto*

- Seismic characterization of stratigraphic sequences.
- Salt bodies and mobile shales.
- Seismic interpretation in structurally complex areas.

8. Seismic imaging workflow and geological structures

- Tomography-based workflow. – *Dr. Cabrera*
- FWI-based workflow. – *Dr. Jones*
- Structural constraints in FWI and ray-based tomography. – *Dr. Jones, Dr. Cabrera.*
- Imaging in and around salt bodies – *Dr. Jones*

9. Constraining seismic images - *Dr. Jones, Prof. Soto, Dr. Cabrera*

- Imaging depth calibration and post-migration depth-to-depth conversion.
- Impact of data uncertainties on imaging results.
- Imaging scenarios for uncertainty assessment.

10. Wrap-up questions and discussion. *Dr. Cabrera, Dr. Jones, Prof. Soto.*